Barren Plateaus: Global vs. Local Cost Functions

Initial Project Synopsis — AI 631 Quantum Machine Learning, Fall 2026 (Project #8, QNN track)

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I. Research Question

Parameterized quantum circuits (PQCs) trained by gradient descent can suffer from *barren plateaus*: the variance of the cost gradient over random parameters vanishes exponentially with the number of qubits n, so gradients become too small to estimate with a polynomial number of shots [1], [2]. Cerezo *et al.* [3] showed that this depends on the cost function: for alternating layered ansatzes, a *global* cost has exponentially vanishing gradients at any depth, while a *local* cost has at worst polynomially vanishing gradients when the depth is $L = \mathcal{O}(\log n)$. The link between cost locality and barren plateaus was also studied in [4]. We ask:

RQ. How does the gradient variance scale with qubit count n and depth L for global and local costs on a hardware-efficient ansatz, does a smarter initialization (identity-block [5] or reduced-domain) help, and how do these effects show up when training a small classifier, in simulation, under a noise model, and on IBM hardware?

II. Background and Related Work

For deep circuits, the gradient variance has been characterized exactly through the circuit's dynamical Lie algebra [6], [7]. Zhang *et al.* [8] gave a general lower bound on the gradient variance for circuits built from local 2-designs. Letcher *et al.* [9] derived tighter gradient bounds without t-design assumptions that can be estimated efficiently on classical hardware, and argued that observables with a non-vanishing local term can avoid barren plateaus. The same global/local split appears in quantum generative models: the MMD loss acts as an observable that is low-bodied and trainable, or global and untrainable, depending on the kernel [10]. Cerezo *et al.* [11] ask whether provable absence of barren plateaus implies classical simulability.

Initialization. Wang *et al.* [12] prove that shrinking each parameter's initial domain in proportion to $1/\sqrt{L}$ makes the gradient decay at most polynomially in n and L. Classical initializers (Xavier, He, LeCun, orthogonal) reduce gradient-variance decay compared with uniform random initialization [13], and the range of the initial distribution alone can mitigate barren plateaus [14].

Noise. Unital noise causes noise-induced barren plateaus [15], and this analysis has been extended to a class of non-unital maps that includes amplitude damping [16]. Mele *et al.* [17] show that noise truncates most circuits to effectively logarithmic depth, and prove that under non-unital noise, costs built from local observables do not have barren plateaus. Engineered dissipation has also been proposed as a way to restore trainability [18].

III. Hypotheses and Falsifiers

H1 (cost locality). At $L = \lceil \log_2 n \rceil$, $\mathrm{Var}[\partial_k C_G]$ decays exponentially in n while $\mathrm{Var}[\partial_k C_L]$ decays much more slowly. *Falsified if*, over $n = 4, \ldots, 14$, the fitted exponential decay rate b (from $\log \mathrm{Var} = a - bn$) of the local cost is not smaller than that of the global cost (overlapping 95% bootstrap CIs), or if the two variances at $n = 14$ differ by less than a factor of 10.

H2 (depth). For the local cost, increasing L from $\lceil \log_2 n \rceil$ to $L \propto n$ makes its variance decay faster with n. *Falsified if* the fitted decay rate does not increase with L.

H3 (initialization). Identity-block or reduced-domain initialization raises the initial gradient variance of the global cost and speeds up training. *Falsified if*, at $n \geq 10$, neither scheme raises the initial gradient variance by at least $10\times$ over uniform initialization, or neither reduces the median number of iterations needed to reach a fixed target loss (10 seeds).

H4 (learning). In the classification task, a model trained through a global readout learns more slowly than the same circuit trained through a local readout, and the gap widens with n. *Falsified if* the test-accuracy curves (mean over seeds) are indistinguishable within one standard deviation at the largest n tested. Negative results will be reported as such.

IV. Models and Costs

Ansatz. Hardware-efficient layered circuit $V(\boldsymbol{\theta})$ on n qubits: each layer applies $R_Y(\theta)R_Z(\theta)$ to every qubit, then CZ gates on pairs $(1,2),(3,4),\ldots$ followed by $(2,3),(4,5),\ldots$ (brickwork). This gives $2nL$ parameters and $(n-1)L$ CZ gates. This ansatz is not guaranteed to satisfy the assumptions in [3], so this is an empirical test.

Costs. With $\rho = V(\boldsymbol{\theta})\,|\psi_0\rangle\langle\psi_0|\,V(\boldsymbol{\theta})^\dagger$,

$$C_G = 1 - \langle 0^{\otimes n}|\rho|0^{\otimes n}\rangle, \tag{1}$$

$$C_L = 1 - \tfrac{1}{n}\textstyle\sum_{j=1}^{n}\langle 0|\rho_j|0\rangle, \tag{2}$$

where ρ_j is the reduced state of qubit j, as defined in [3]. Both costs come from the same Z-basis measurement: C_G uses the all-zeros frequency and C_L uses the single-qubit marginals.

Sanity check. For the product ansatz $\bigotimes_j R_X(\theta_j)$ on $|0\rangle^{\otimes n}$ with $\theta_j \sim U[0, 2\pi]$, a direct calculation gives $\mathrm{Var}[\partial_1 C_G] = \frac{1}{8}(\frac{3}{8})^{n-1}$ and $\mathrm{Var}[\partial_1 C_L] = \frac{1}{8n^2}$. We will use this closed form as a unit test for the variance-estimation code.

Initializations. (i) Uniform $\theta \sim U[0, 2\pi]$; (ii) reduced-domain $\theta \sim U[-a, a]$ with $a \propto 1/\sqrt{L}$, following [12]; (iii)

identity blocks [5], where each block is a random sub-circuit followed by its adjoint, so the block starts as the identity.

Training task. A fixed random shallow circuit W prepares $|\psi_0\rangle = W|0\rangle$, and $V(\boldsymbol{\theta})$ is trained to map it back to $|0\rangle^{\otimes n}$. This keeps identity initialization from being trivially optimal.

Classifier. Angle encoding of n PCA features, followed by the same ansatz. The class score is either global ($P(0^{\otimes n})$) or local ($\frac{1}{2}(1 + \frac{1}{n}\sum_j \langle Z_j \rangle)$), trained with binary cross-entropy. Gradients are exact on the statevector simulator and use the parameter-shift rule [19] with finite shots and on hardware.

V. Data and Baselines

Gradient-variance studies use random circuits only, so no external data is needed. The classification task uses the Breast Cancer Wisconsin (Diagnostic) dataset shipped with scikit-learn, reduced by PCA to n features ($n \in \{4, 6, 8, 10\}$). We use the course NB11 starter harness for stratified folds and leakage-free preprocessing (scaler and PCA fit on the training fold only), with fixed seeds. The classical baselines are majority class, logistic regression, and an MLP with a parameter count matched to the quantum model. We report MLP training curves on the same folds next to the quantum curves. For gradient scaling, the reference is the analytic product-ansatz result above.

VI. Experimental Protocol and Resources

Noise-free sweeps: $n \in \{2, 4, \ldots, 14\}$, $L \in \{1, \lceil \log_2 n \rceil, n\}$, 200 parameter draws per $(n, L, \text{cost}, \text{init})$. We report the variance of $\partial_k C$ for a fixed k and averaged over k, with bootstrap CIs. **Noisy simulation:** Qiskit Aer with depolarizing (unital) and amplitude-damping (non-unital) channels, plus a noise model taken from an IBM fake backend. This separates cost-induced from noise-induced plateaus [15], [16] and tests whether the local cost holds up better under non-unital noise, as [17] suggests. **Hardware:** parameter-shift gradients on an IBM backend at $n \in \{2, 4, 6\}$, $L = \lceil \log_2 n \rceil$, 30–50 random draws, 2 circuits per gradient, 4096 shots, so at most 300 circuits in total, with both costs from the same circuits. We report the shot-noise floor next to measured variances. Software: PennyLane and Qiskit [20].

VII. Week-by-Week Plan

Week (due)	Work
0 (Nov 14)	Repo and environment; analytic unit test; ansatz and both costs; variance-estimation script.
1 (Nov 15)	Noise-free variance sweep for $n \leq 14$, uniform initialization; classifier pipeline with NB11 folds; classical baselines; first quantum classifier. Report: introduction and methods.
2 (Nov 22)	Full (n, L) sweeps; three initializations; training task; global vs. local classifier curves; noise-model runs and ablations. Report: results draft.
3 (Nov 29)	IBM hardware parameter-shift gradients at $n = 2, 4, 6$; compare with the noise model; presentation if scheduled.
4 (Dec 6)	Final IEEE report (PDF and source), final talk, tagged GitHub release with saved results and scripts to reproduce every figure.

References

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